Dynamic Pricing Competition: Booking Curves

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Dynamic pricing competition:

Competition setup:

- 80 tickets can be sold in 100 time steps: on average 0.8 tickets per step
- revenue achieved: $\sum_{t \in T} p_t d_t$ where p_t is your price level and d_t is the served demand in time period t
- In the current time period $t_{current,}$ you know the revenue you achieved until $t_{current} - 1$ and you can estimate if your demand is larger, smaller, or comparable to the served demand of your competitor
- The demand curve is unknown but you know it is monotonously increasing in time and decreasing in price
- Performance indicators:
  - Your current revenue vs. the estimated competitor's revenue
  - The number of tickets sold: you vs. your competitor

Dynamic pricing competition: Booking curves

Idea:

- Plan in advance how many tickets you want to sell until some predefined time steps, e.g. T/4, T/2, ¾ T, …
  - Maybe non-uniform time intervals are better?
- This gives you a booking curve over time (time intervals)
- Crosscheck whether you have sold as many tickets as you preplanned
  - Increase price level if you have sold already too many tickets
  - If you are performing as planned, keep price level
  - Decrease price level, if there are less tickets sold than planned
- Try to minimize the deviation from your preplanned booking curve
- Use suitable threshold values and margins

Dynamic pricing competition: Revenue curves

Idea:

- Monitor your current revenue and the estimated revenue and inventory level of your competitor
  - Try to keep your target revenue above the final estimate of your competitor´s revenue
  - If there is a (good) chance that your competitor runs out of stock:
    - estimate if you can still outperform its revenue